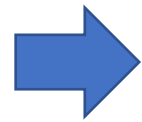


4-4

Coordinate System

- Let vector set $\mathcal{B} = \{u_1, u_2, \dots, u_n\}$ be a ^{ind.} **basis** for a subspace $\mathbb{R}^n$



$\mathcal{B}$ is a coordinate system

- For any v in $\mathbb{R}^n$, there are unique scalars $c_1, c_2, \dots, c_n$ such that $v = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$

$\mathcal{B}$ -coordinate vector of v :

$$[v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \in \mathcal{R}^n$$

(用 $\mathcal{B}$ 的觀點來看 v)

95. Let $A = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for R^n and $c_1, c_2, \dots, c_n$ be nonzero scalars. Recall that $B = \{c_1\mathbf{u}_1, c_2\mathbf{u}_2, \dots, c_n\mathbf{u}_n\}$ is also a basis for R^n . If $\mathbf{v}$ is a vector in R^n and

$$[\mathbf{v}]_A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Compute $[\mathbf{v}]_B$.

$$[\mathbf{v}]_A \xrightarrow{\textcircled{1}} \mathbf{v} \xrightarrow{\textcircled{2}} [\mathbf{v}]_B$$

$$\textcircled{1} \quad \mathbf{v} = a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \dots + a_n\mathbf{u}_n$$

$$\textcircled{2} \quad \mathbf{v} = \underbrace{?}_{\frac{a_1}{c_1}} (c_1\mathbf{u}_1) + \underbrace{?}_{\frac{a_2}{c_2}} (c_2\mathbf{u}_2) + \dots + \underbrace{?}_{\frac{a_n}{c_n}} (c_n\mathbf{u}_n)$$

$$= \begin{bmatrix} \frac{a_1}{c_1} \\ \frac{a_2}{c_2} \\ \vdots \\ \frac{a_n}{c_n} \end{bmatrix}$$

97. Let $A = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for R^n and $c_1, c_2, \dots, c_n$ be nonzero scalars. Recall that $B = \{\mathbf{u}_1 + \mathbf{u}_2 + \dots + \mathbf{u}_n, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is also a basis for R^n . If $\mathbf{v}$ is a vector in R^n and

$$[\mathbf{v}]_A \xrightarrow{\textcircled{1}} \mathbf{v} \xrightarrow{\textcircled{2}} [\mathbf{v}]_B$$

$$[\mathbf{v}]_A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Compute $[\mathbf{v}]_B$.

$$\textcircled{1} \quad \mathbf{v} = \underline{a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_n \mathbf{u}_n}$$

$$\textcircled{2} \quad \mathbf{v} = \boxed{a_1} (\mathbf{u}_1 + \mathbf{u}_2 + \dots + \mathbf{u}_n) + \boxed{a_2 - a_1} \mathbf{u}_2 + \dots + \boxed{a_n - a_1} \mathbf{u}_n$$

$$= \begin{bmatrix} a_1 \\ a_2 - a_1 \\ \vdots \\ a_n - a_1 \end{bmatrix}$$

102. Let $\underline{A}$ and $\underline{B}$ be two bases for R^n . Express $\underline{[v]}_A$ in terms of $\underline{[v]}_B$.

$\underline{A}$ $\underline{B}$

$$\underline{[v]}_A \xrightarrow{\textcircled{1}} v \longrightarrow \underline{[v]}_B$$

$$\textcircled{1} \quad v = A \underline{[v]}_A$$

$$\textcircled{2} \quad \underline{[v]}_B = B^{-1} v$$

$$v = B \underline{[v]}_B$$

$$\underline{[v]}_B = B^{-1} A \underline{[v]}_A$$

$$\underline{[v]}_A = A^{-1} B \underline{[v]}_B$$